

MOUNTING USB WITH THE NTFS TYPE

OBJECTIVE

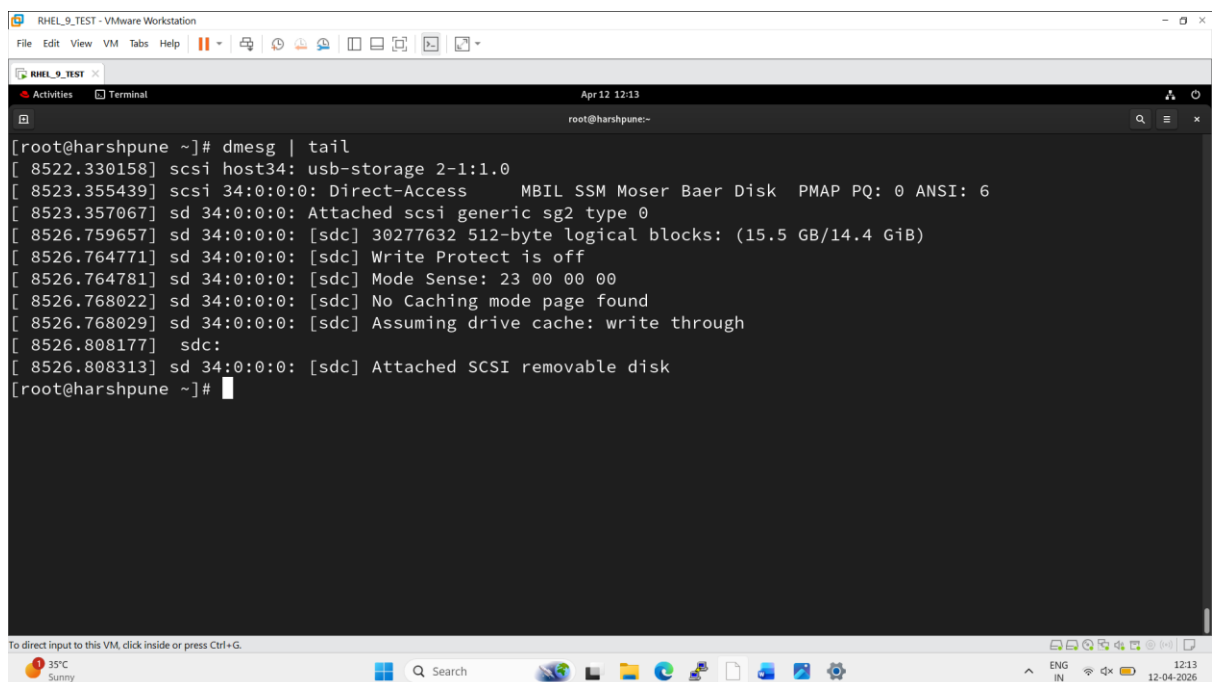
THIS DOCUMENT EXPLAINS HOW TO :

- FORMAT A USB DRIVE WITH NTFS FILESYSTEM
- MOUNT THE USB DRIVE IN LINUX

STEP 1 : IDENTIFY THE USB DRIVE

- INSERT THE USB AND RUN

`dmesg | tail`



```
[root@harshpune ~]# dmesg | tail
[ 8522.330158] scsi host34: usb-storage 2-1:1.0
[ 8523.355439] scsi 34:0:0:0: Direct-Access    MBIL SSM Moser Baer Disk  PMAP PQ: 0 ANSI: 6
[ 8523.357067] sd 34:0:0:0: Attached scsi generic sg2 type 0
[ 8526.759657] sd 34:0:0:0: [sdc] 30277632 512-byte logical blocks: (15.5 GB/14.4 GiB)
[ 8526.764771] sd 34:0:0:0: [sdc] Write Protect is off
[ 8526.764781] sd 34:0:0:0: [sdc] Mode Sense: 23 00 00 00
[ 8526.768022] sd 34:0:0:0: [sdc] No Caching mode page found
[ 8526.768029] sd 34:0:0:0: [sdc] Assuming drive cache: write through
[ 8526.808177]   sdc:
[ 8526.808313] sd 34:0:0:0: [sdc] Attached SCSI removable disk
[root@harshpune ~]#
```

- NOW CHECK IN

`lsblk | grep disk`

STEP 2 : REMOVE THE EXISTING FILE SYSTEM IF ANY

- `lsblk -f`

```
[root@harshpune ~]# lsblk | grep disk
sda      8:0      0    5G  0 disk
sdc      8:32     1 14.4G  0 disk
nvme0n1 259:0     0   20G  0 disk
```

- CHECK FOR ANY FILESYSTEM ALREADY EXISTS

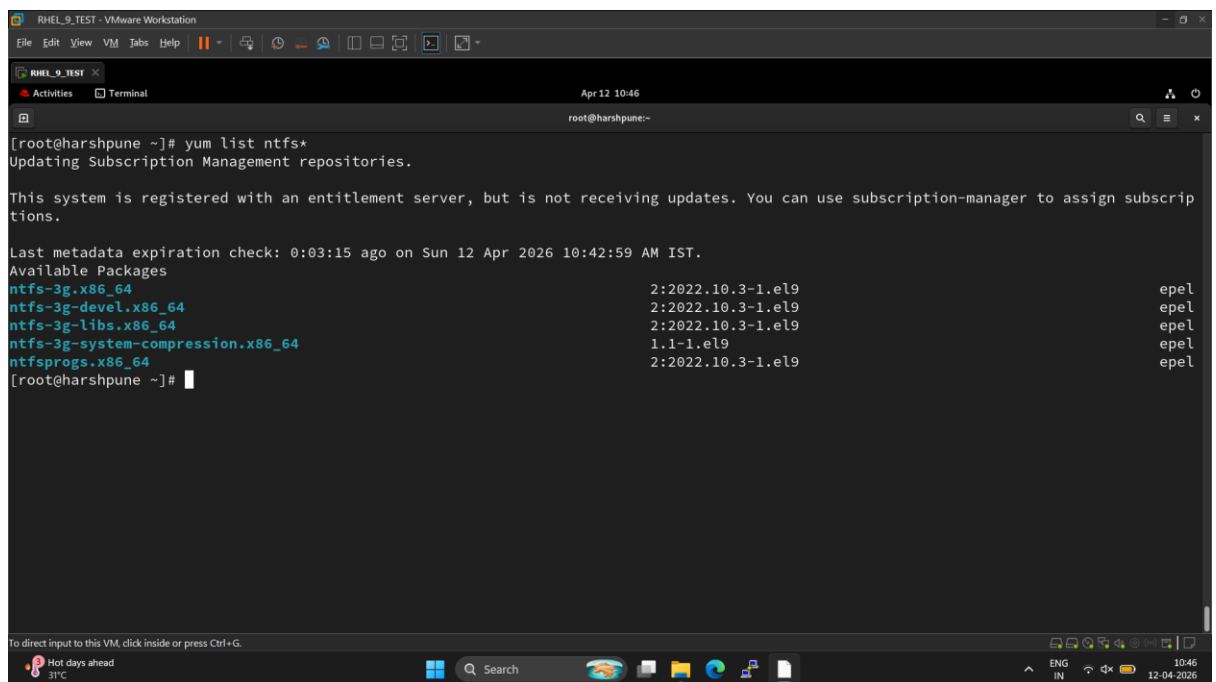
```
[root@harshpune ~]# lsblk -f
NAME        FSTYPE FSVER LABEL                                UUID                                  FSAVAIL FSUSE% MOUNTPOINTS
├── sda
│   ├── sda1
│   ├── sda2
│   └── sda3
├── sdc      ext4    1.0
│   └── sr0   iso9660 Jolie RHEL-9-6-0-BaseOS-x86_64      73cd3269-d8a2-4e8d-bda8-e94faff42de8 0      100% /mnt
│                                               /run/media/root/RHEL-9-6-0-BaseOS-x86_64
└── nvme0n1
    ├── nvme0n1p1 vfat    FAT32
    │   └── nvme0n1p2 xfs
    │       └── nvme0n1p3 swap     1
    └── nvme0n1p3 swap     1

NAME        FSTYPE FSVER LABEL                                UUID                                  FSAVAIL FSUSE% MOUNTPOINTS
├── sda
│   ├── sda1
│   ├── sda2
│   └── sda3
├── sdc      ext4    1.0
│   └── sr0   iso9660 Jolie RHEL-9-6-0-BaseOS-x86_64      73cd3269-d8a2-4e8d-bda8-e94faff42de8 0      100% /mnt
│                                               /run/media/root/RHEL-9-6-0-BaseOS-x86_64
└── nvme0n1
    ├── nvme0n1p1 vfat    FAT32
    │   └── nvme0n1p2 xfs
    │       └── nvme0n1p3 swap     1
    └── nvme0n1p3 swap     1
```

- IN MY CASE I HAVE EXT4 FILESYSTEM
- SO NOW WE NEED TO REMOVE THE EXISTING FILESYSTEM
- NOW RUN THE COMMAND
`wipefs -a /dev/sdc`
- NOW WE WILL GET DISK WITHOUT ANY FILESYSTEM

STEP 3 : FORMAT THE BLANK DISK WITH NTFS FILESYSTEM

- AS WE KNOW THAT NTFS IS BY DEFAULT NOT SUPPORTED IN LINUX SO WE NEED TO INSTALL SOME PACKAGES
- NOW RUN THE COMMAND TO LIST THE REQUIRED PACKAGES
- `yum install ntfs*`



```
[root@harshpune ~]# yum list ntfs*
Updating Subscription Management repositories.

This system is registered with an entitlement server, but is not receiving updates. You can use subscription-manager to assign subscriptions.

Last metadata expiration check: 0:03:15 ago on Sun 12 Apr 2026 10:42:59 AM IST.
Available Packages
ntfs-3g.x86_64                2:2022.10.3-1.el9          epel
ntfs-3g-devel.x86_64        2:2022.10.3-1.el9          epel
ntfs-3g-libs.x86_64         2:2022.10.3-1.el9          epel
ntfs-3g-system-compression.x86_64 1:1-1.el9                  epel
ntfsprogs.x86_64            2:2022.10.3-1.el9          epel
[root@harshpune ~]#
```

- SO WE NEED TO INSTALL THE FOLLOWING TO PACKAGES

`ntfs-3g.x86_64`

`ntfsprogs.x86_64`

- INSTALL THE PACKAGES

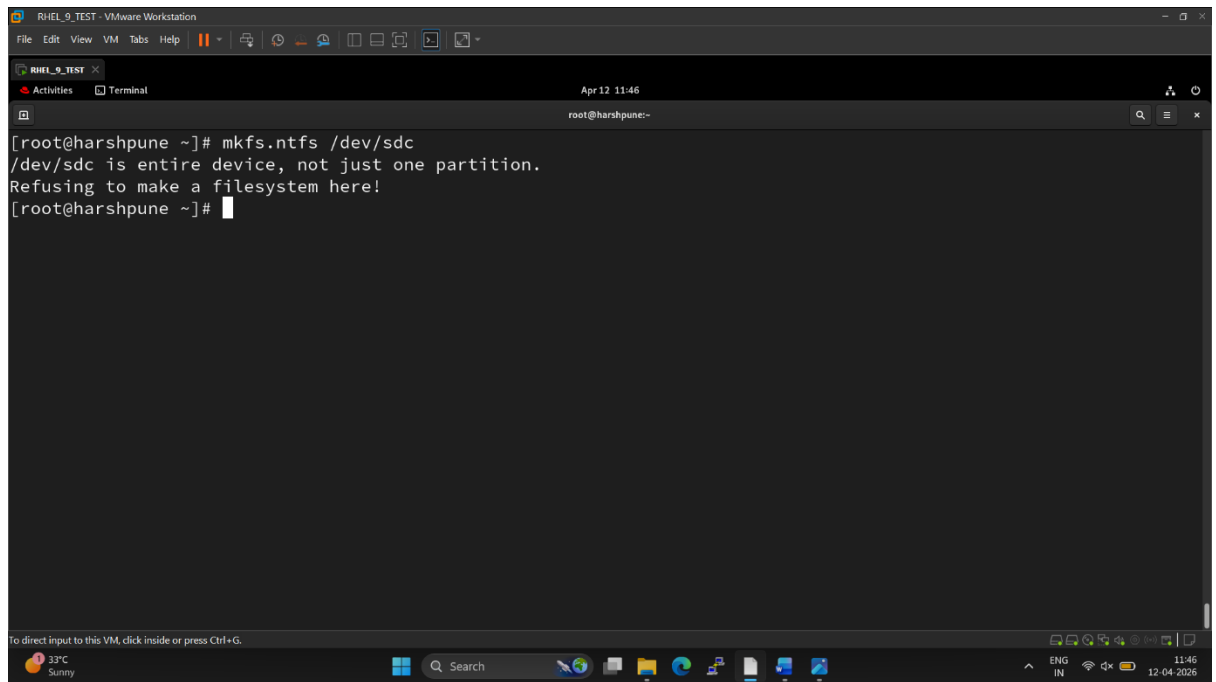
`yum install ntfs-3g.x86_64 -y`

`yum install ntfsprogs.x86_64 -y`

- NOW MAKE FILE SYSTEM IN THE USB DISK

`mkfs.ntfs /dev/sdc`

- AN ERROR MIGHT APPEAR AS

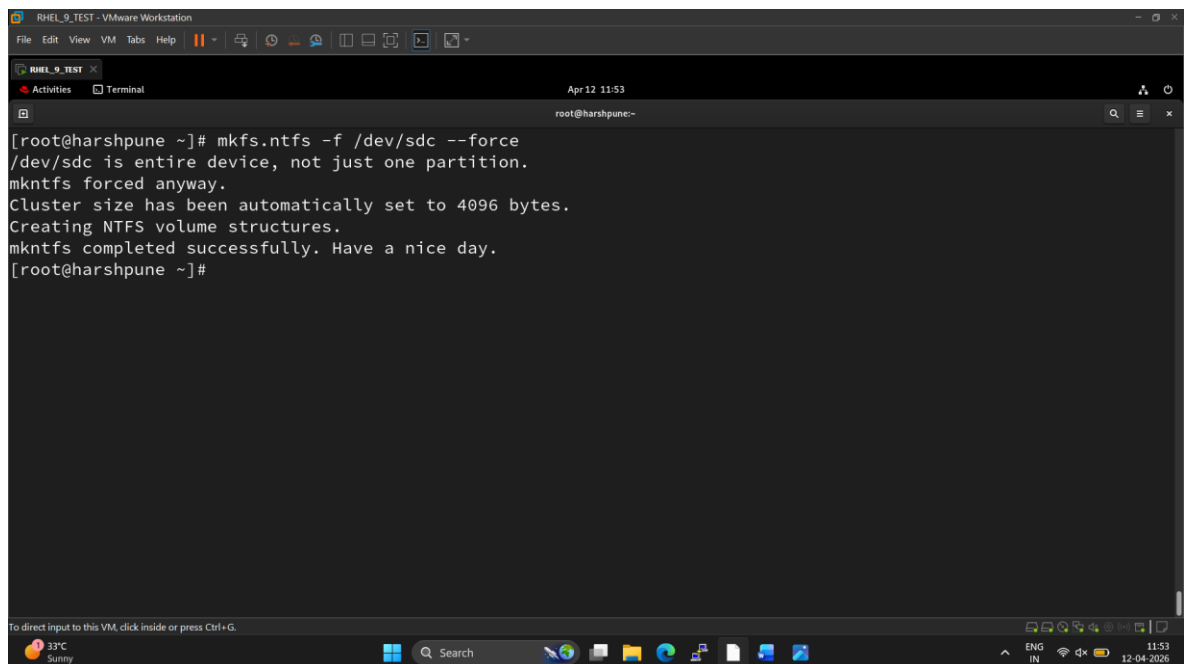


The screenshot shows a terminal window titled "RHEL_9_TEST" within a VMware Workstation. The terminal output is as follows:

```
[root@harshpune ~]# mkfs.ntfs /dev/sdc
/dev/sdc is entire device, not just one partition.
Refusing to make a filesystem here!
[root@harshpune ~]#
```

The terminal window has a menu bar (File, Edit, View, VM, Tabs, Help) and a toolbar. The bottom status bar shows the date and time as "Apr 12 11:46" and the user as "root@harshpune:-".

- NOW WE CAN FORCE THE SYSTEM TO FORMAT IT WITH USB BY
`mkfs.ntfs -f /dev/sdc --force`
here -f stands for faster
--force stands for force to make it
- NOW THE DISK IS DONE FORMATTING WITH NTFS FILESYSTEM

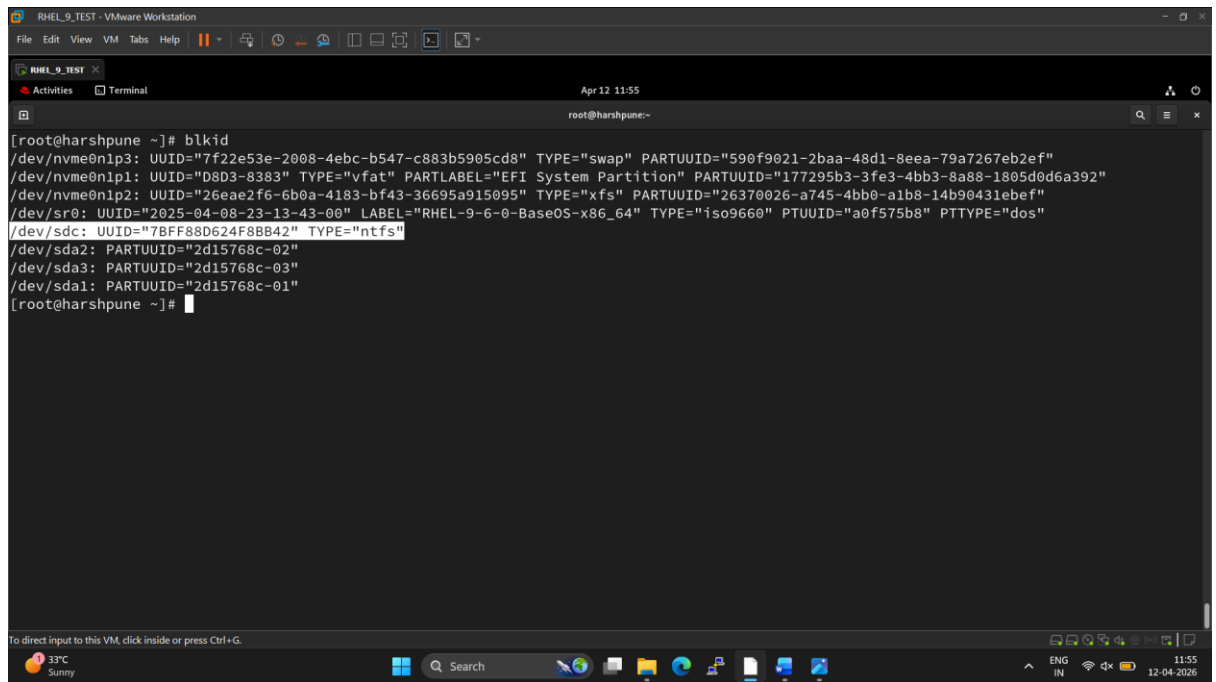


The screenshot shows the same terminal window after running the command with the -f and --force flags. The output is:

```
[root@harshpune ~]# mkfs.ntfs -f /dev/sdc --force
/dev/sdc is entire device, not just one partition.
mkntfs forced anyway.
Cluster size has been automatically set to 4096 bytes.
Creating NTFS volume structures.
mkntfs completed successfully. Have a nice day.
[root@harshpune ~]#
```

The terminal window's status bar now shows the date and time as "Apr 12 11:53".

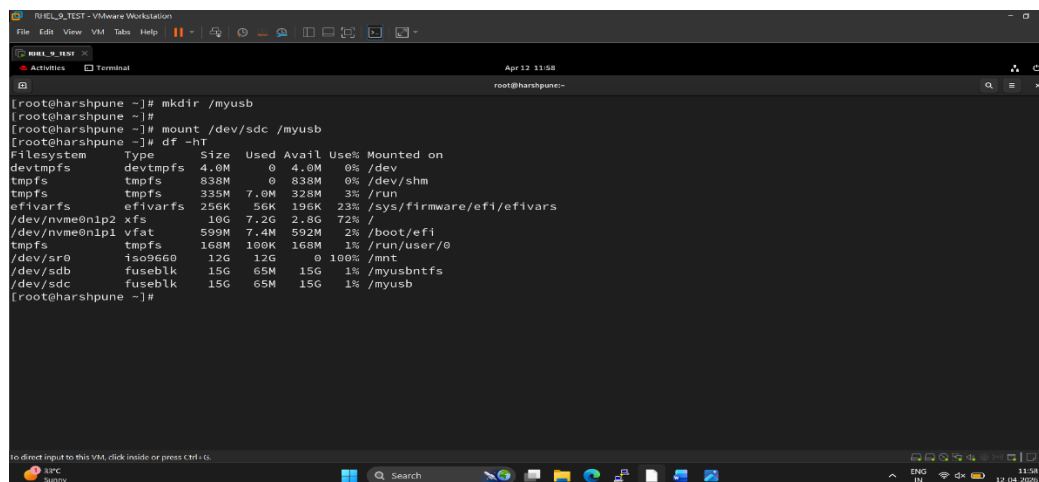
- WE CAN ALSO CONFIRM THIS IN blkid



```
[root@harshpune ~]# blkid
/dev/nvme0n1p3: UUID="7f22e53e-2008-4ebc-b547-c883b5905cd8" TYPE="swap" PARTUUID="590f9021-2baa-48d1-8eea-79a7267eb2ef"
/dev/nvme0n1p1: UUID="D8D3-8383" TYPE="vfat" PARTLABEL="EFI System Partition" PARTUUID="177295b3-3fe3-4bb3-8a88-1805d0d6a392"
/dev/nvme0n1p2: UUID="26eae2f6-6b0a-4183-bf43-36695a915095" TYPE="xfs" PARTUUID="26370026-a745-4bb0-a1b8-14b90431ebef"
/dev/sr0: UUID="2025-04-08-23-13-43-00" LABEL="RHEL-9-6-0-BaseOS-x86_64" TYPE="iso9660" PTUID="a0f575b8" PTTYPE="dos"
/dev/sdc: UUID="7BFF88D624F8BB42" TYPE="ntfs"
/dev/sda2: PARTUUID="2d15768c-02"
/dev/sda3: PARTUUID="2d15768c-03"
/dev/sda1: PARTUUID="2d15768c-01"
[root@harshpune ~]#
```

STEP 4 : MOUNT THE DISK

- MAKE AN EMPTY DIRECTORY FOR MOUNT POINT
`mkdir /myusb`
- NOW MOUNT THE DISK
`mount /dev/sdc /myusb`
- CHECK IN `df -hT`



```
[root@harshpune ~]# mkdir /myusb
[root@harshpune ~]# mount /dev/sdc /myusb
[root@harshpune ~]# df -hT
Filesystem      Type      Size  Used Avail Use% Mounted on
devtmpfs        devtmpfs  4.0M   0  4.0M   0% /dev
tmpfs           tmpfs     838M   0  838M   0% /dev/shm
tmpfs           tmpfs     335M   7.0M  328M   3% /run
efivarfs        efivarfs  256K   56K  196K  23% /sys/firmware/efi/efivars
/dev/nvme0n1p2  xfs       10G   7.2G  2.8G  72% /
/dev/nvme0n1p1  vfat      599M   7.4M  592M   2% /boot/efi
tmpfs           tmpfs     168M  190K  168M   1% /run/user/0
/dev/sr0        iso9660   12G   12G   0 100% /mnt
/dev/sdb        fuseblk   15G   65M   15G   1% /myusbntfs
/dev/sdc        fuseblk   15G   65M   15G   1% /myusb
[root@harshpune ~]#
```